Lynq

Lynq NTN-Pulse

Every signal, the right path.

Digital twin + policy engine for OQ's regenerative gNB-onboard LEO constellation

AnyPoint-NTN challenge · Next-Gen Downstream · NEOSPACE Europe 2026, Luxembourg

Davoud Jokar · Ines Bouissou · Gambhir Sharma · Maryam Amini

Bent-pipe satellites fail where offshore assets need them



No gateway, no service

A bent-pipe satellite only relays. If no ground gateway is in view at the same time as the terminal, the message waits or drops.



Gateway not allowed

A gateway can be in view but blocked by national rules, data residency or licensing. Technically fine, legally unusable.



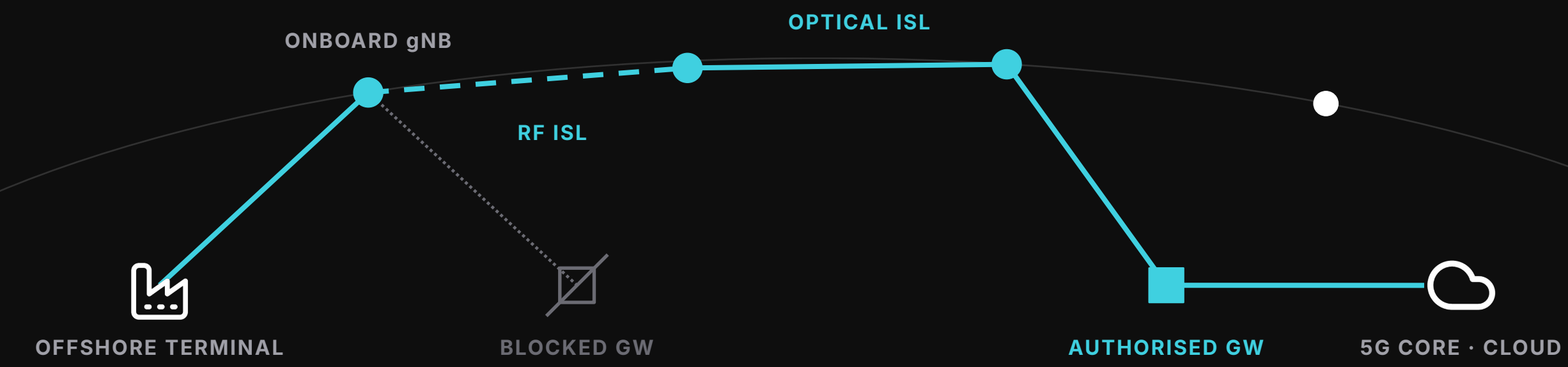
Mixed traffic, one queue

A shutdown alarm and a 100 kB log file get the same treatment. Critical traffic should never wait behind bulk data.



Who is hit hardest: offshore wind farms, oil & gas platforms, vessels and pipelines in the North Sea (~51–61°N), far from terrestrial cells, across five national sectors (UK, NO, DK, NL, DE).

NTN-Pulse picks a compliant path for every message



CRITICAL

Alarms and shutdown commands. First uplink slot, first authorised path, this pass.

REGULAR

Telemetry, logs, files. Queued onboard and store-and-forwarded on a later pass (≤ 90 min).

METADATA

Status, position, policy tags. Tells the engine which gateway each asset may use.

Regenerative gNB on every satellite, RF + optical mesh

~12 → 30

satellites: 10 OQ + 2 ERMIS today, 30 planned

3 × 10

planes × satellites at ~500 km altitude

10°

minimum elevation, ~3,000 km footprint

~6 / yr

replacements at a 5-year design life

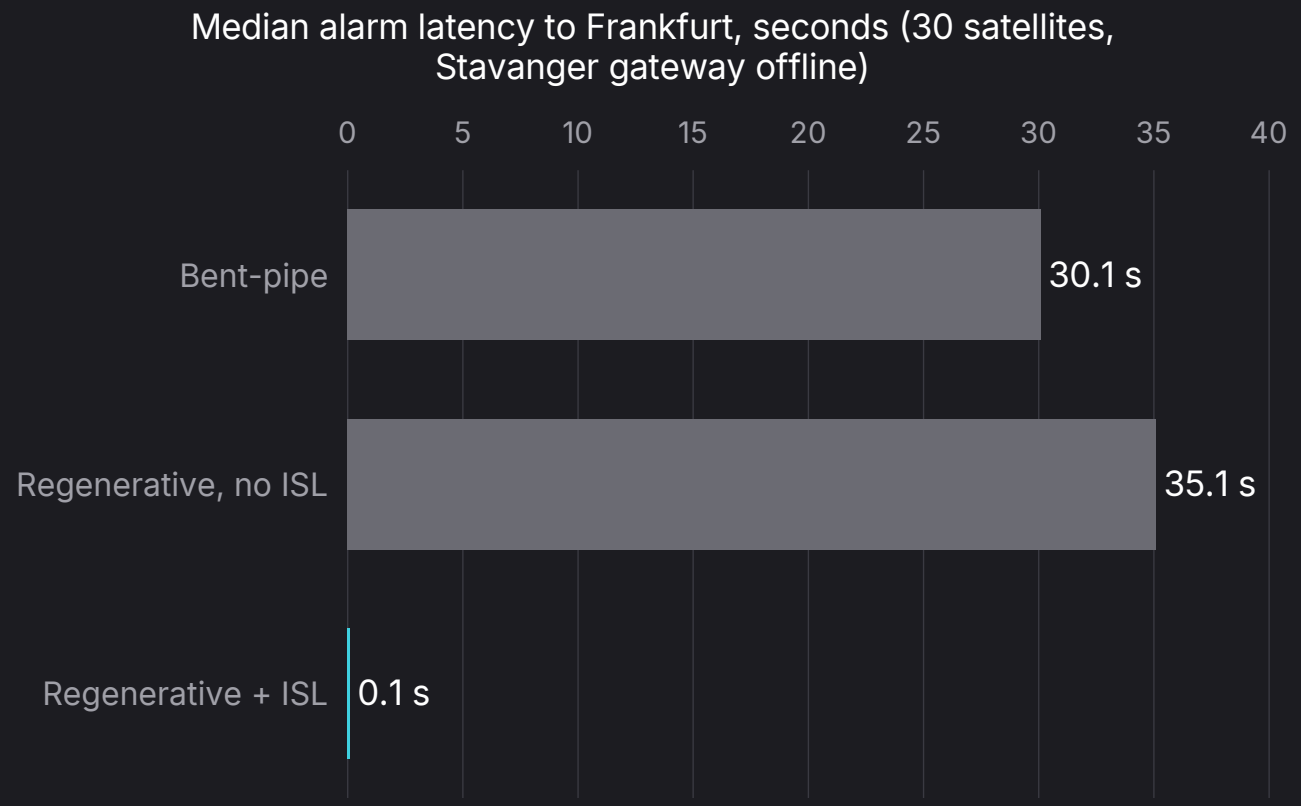
Payload: full regenerative gNB

Bent-pipe: needs a gateway in view; can't serve open ocean
Split gNB (2 / 7.2x): RRC still round-trips the feeder
SWaP fix: narrowband gNB, duty-cycling, rad-hard supervisor

ISL: RF in-plane, optical cross-plane

RF, every satellite: static in-plane geometry, proven by Iridium
Optical, relay subset: no licence, EU suppliers, no ITAR
Not all-optical: IoT needs only single-digit Mbps per sat

ISL keeps alarms at ~100 ms even with a gateway down



~85 ms

live alarm, platform → Frankfurt cloud (30 sats, all gateways up)

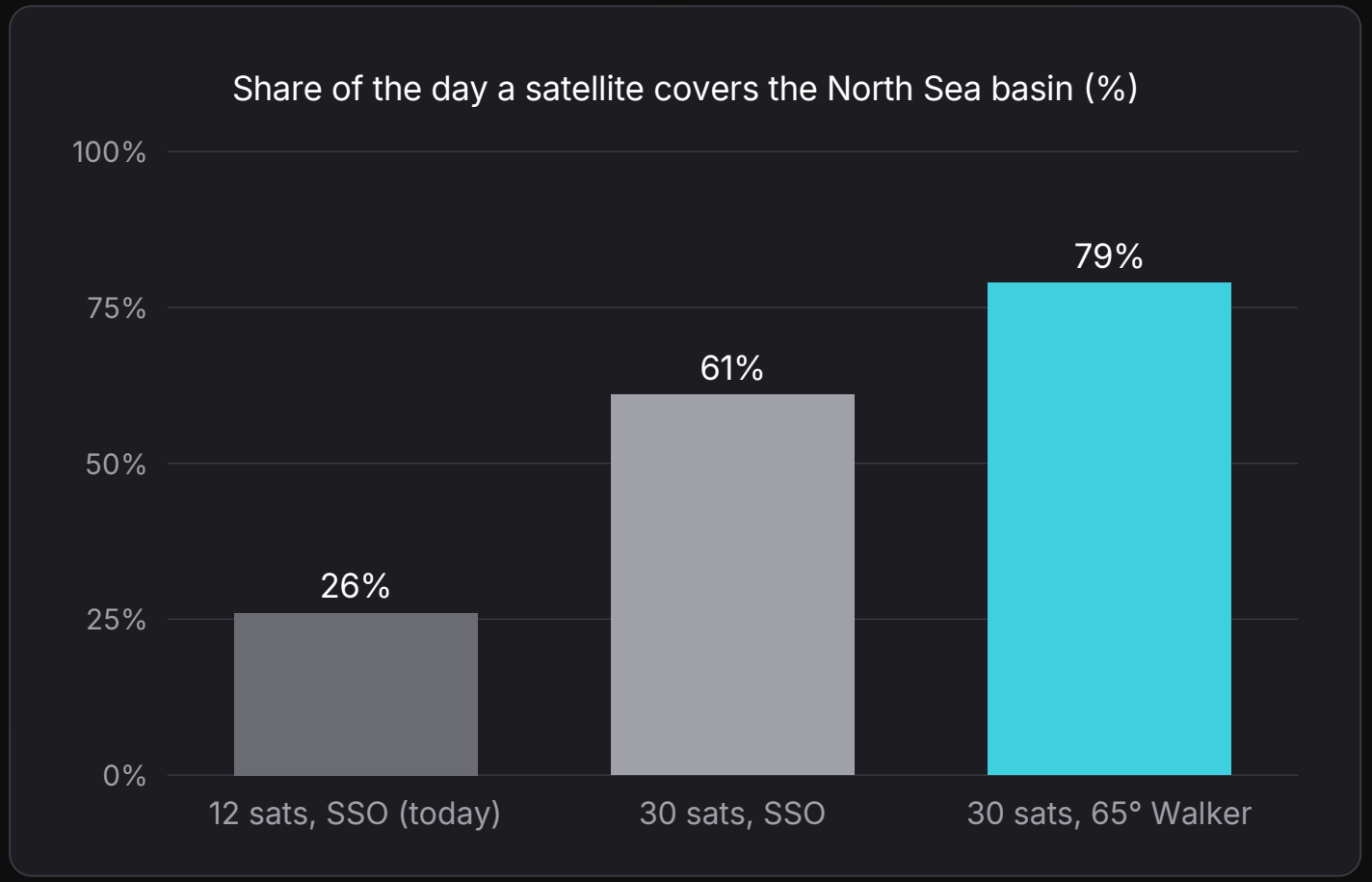
75–110 ms

mid-Atlantic platform → Frankfurt over the ISL mesh; bent-pipe: no service

SERVICE TIERS

Continuous	< 150 ms
Mesh-relayed	< 250 ms
Store-and-forward	≤ 90 min

Orbit choice beats satellite count for the North Sea



LONGEST COVERAGE GAP

246 min — 12 sats today: passes, not coverage

118 min — 30 sats, sun-synchronous planes

3 min — same 30 sats, 65° Walker delta

75% of alarms under 250 ms with the Walker delta (vs 59% on sun-synchronous)

Sovereignty is built into routing, starting in Luxembourg

1

Luxembourg ILR / LSA

Extend OQ's Nov-2025 concession to ISL and regenerative ops

2

ITU BR via Luxembourg

Priority filings: Art. 9 coordination, Art. 22 EPFD

3

International waters

No landing rights needed: earliest revenue

4

North Sea regulators + MNOs

UK, NO, DK, NL, DE for coastal coverage

5

EC / CEPT

S-band MSS (IoT) and C-band D2D spectrum

Data residency, lawful intercept

The gNB knows identity, PLMN and verified location, so each flow lands only at an allowed gateway with local breakout. MNO keeps lawful intercept. 100% compliant in the twin.

Spectrum as a hard constraint

ITU Art. 9 / Art. 22 EPFD limits, feeder-link masks and gateway exclusion zones are inputs to routing, not afterthoughts.

Export control and debris

European optical ISL and gNB suppliers keep us outside ITAR/EAR. 5-year life, ~6 replacements a year, IADC deorbit compliance.

Start with North Sea offshore energy, priced per asset

First buyer

European offshore energy operator or wind O&M provider (e.g. Equinor, Ørsted, Eni)

Pain: blind spots, missed alarms, costly VSAT and fixed gateways

Why first: international waters, sovereign-minded buyers, OQ's European funding focus

BREAK-EVEN

~€9M / yr

opex for the North Sea network, incl. replenishment

~75k assets
× €120 / yr IoT subscription
or
~9k endpoints
× €1,000+ / yr sovereign / defence

REVENUE MODEL

- **Integration fee**
  per customer onboarding
- **Subscription**
  per asset and traffic class
- **Priority QoS**
  usage charge for critical tier
- **MNO wholesale**
  roaming for coastal coverage

Per asset, never per byte: at ~€2–3/MB, byte pricing would kill adoption.

We can't out-build the giants, so we out-position them

APPROACH	EXAMPLE	LATENCY / COVERAGE	SOVEREIGNTY / COST
Bent-pipe NTN IoT	Sateliot (~40 LEO)	Waits for a gateway in view; store-and-forward	EU, but Rel-17 bent-pipe only
GEO NTN IoT	Skylo	~240 ms round trip, no polar coverage	Rents someone else's satellites
Regenerative direct-to-cell	Starlink D2C, AST	Low latency, huge ISL mesh	US-controlled, billions in capex
Proprietary RF mesh	Iridium (66 sats)	True global, proven since the 1990s	Proprietary, never cheap
Lynq NTN-Pulse on OQ	**~12 → 30 regen sats**	**~85–100 ms alarms, 3-min worst gap (65° delta)**	**EU-sovereign routing, 3GPP-open, per-asset**

Our edge: European sovereignty, data residency by design, and open 3GPP standards for buyers who will not route through US networks.

De-risk with one paid pilot: a wind-farm alarm use case

RISK	MITIGATION
Payload Final power, compute, radiation and RAN limits may differ	Payload-independent policies with cloud/gateway fallback
ISL / regulatory ISL, gateway and spectrum choices may change	Simulate both ISL types; ITU and national rules as hard constraints
Adoption Customers may resist new terminals or uncertain SLAs	One alarm use case, measured TCO, a paid pilot with SLA targets



Next steps

- Phase 2 (here): live twin of one offshore corridor at 12 and 30 sats, bent-pipe vs regenerative
- Validate the 65° Walker delta with OQ's launch plan
- Paid pilot with one North Sea operator: latency, availability, delivery targets
- Orbit-aware fleet updates: 1.2 h with ISL vs 21 h without

Lynq

Every signal, the right path.

Thank you. Questions?

~85 ms
alarms to the cloud

3 min
worst North Sea gap

100%
routing compliance